TBXT Hit Identification

Top 4 Picks for Chordoma's Master Regulator

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The headline

137

STRICTLY COMPLIANT COMPOUNDS
DISTILLED FROM A 570-POOL

4

PICKS FOR JUDGES

3.2 μM

BEST PREDICTED BOLTZ KD
(DUAL-ENGINE CROSS-VAL 1.02×)

Target

Human TBXT (Brachyury) · 435 aa · DBD residues 42-219

- **G177D** = `rs2305089`, common chordoma variant in **>90% of Western cases** (allele freq 42%)
- Site F = `Y88 / D177 / L42` — pocket directly engages the variant residue
- Sequence-aware selectivity: **G177 is 0% conserved** across 16 T-box paralogs; M181 7%, T183 13%
- Receptor: **PDB 6F59 chain A** (G177D + DNA — matches CF Labs SPR construct)

> **Why site F:** TEP-recommended · all best Naar SPR binders predicted at F · TBXT-unique residues = intrinsic selectivity

The pipeline — 6 orthogonal signals

Signal	What it catches
Vina ensemble (6 receptor confs)	Geometric fit + receptor flexibility
GNINA CNN pose + pKd	Vina-trap detection (contact-maximizing decoys)
TBXT QSAR (RF + XGBoost on 650 measured Naar SPR Kd)	The **only on-target signal**
Boltz-2 co-folding — Jack local + SCC re-run	Independent affinity + binder/non-binder classifier
MMGBSA (OpenMM + GBn2, top 30)	Free-energy refinement
T-box paralog selectivity (16 paralogs)	Off-target risk

> Each signal has a known failure mode → **caught by another signal in the stack**

The 7-criterion strict filter (T-0 hard gate)

Of 570 pool compounds → 137 pass all 7 → 4 picks (ranks 1, 2, 11, 22)

Organizer-defined hard: - **C1** onepot 100% catalog match (similarity = 1.000) - **C2** strictly non-covalent - **C3** Chordoma rule - **C5** PAINS-clean + no acid halides / aldehydes / diazo / imines / etc. - **C6** Tanimoto < 0.85 to Naar / TEP / prior_art

Lead-like + feasibility: - **C4** 10–30 HA · HBD+HBA ≤ 11 · LogP < 5 · < 5 rings · ≤ 2 fused - **C7** ESOL log S > -5 (DMSO @ 10 mM + aqueous @ 50 µM) **Tier breakdown of 137:** T1 GOLD: 0 (empty by design — honest) · T2 SILVER: 16 · T3 BRONZE: 89 · T4 RELAXED: 32

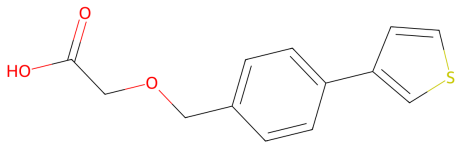
The 4 picks

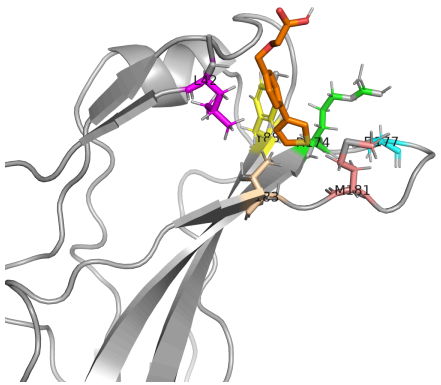
#	ID	Boltz Kd	gnina Vina	$	risks
1	FM002150_analog_0083	**3.2 µM**	-5.01	**$125**	**low/low**
2	FM001452_analog_0104	**3.7 µM**	-5.77	$250	med/med
3	FM001452_analog_0201	8.16 µM	**-6.07**	$375	high/med
4	FM001452_analog_0171	8.32 µM	-6.19	$250	med/med

$875 total to source · **4/4 site F** · **dual-engine Boltz 1.01-1.34×** agreement on every pick

Pick 1 — `FM002150_analog_0083`

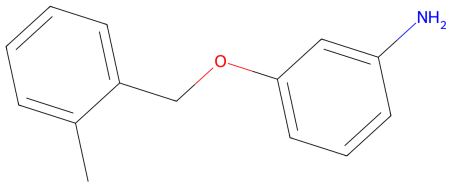
`O=C(O)COCc1ccc(-c2ccsc2)cc1` **Best of all 4 metrics** - **Boltz Kd 3.2 µM** — strongest 100%-onepot non-covalent - Phenoxyacetic acid + thiophene; carboxylate H-bonds **Y88 / D177** - **$125 + low/low risk** — cheapest + lowest-risk pick - MW 248.3 · LogP 3.02 · HBD 1 · HBA 4 - Dual-engine Boltz 1.02× agreement

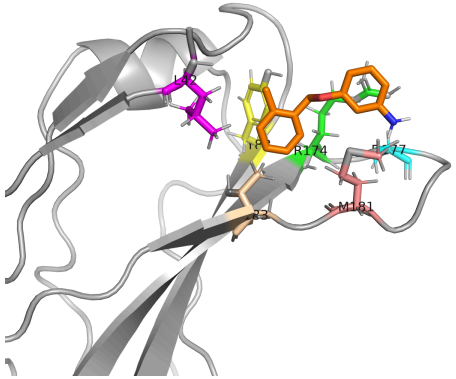




Pick 2 — `FM001452_analog_0104`

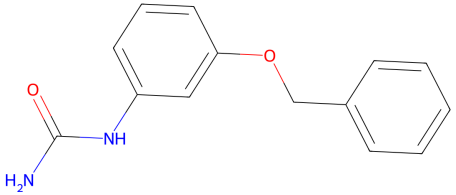
`Cc1ccccc1COc1cccc(N)c1` **Cleanest medchem in the pool** - **Boltz Kd 3.7 / 4.97 μM** — second-strongest - Methyl-phenyl-CH_2-O-aniline; aniline-N H-bonds **D177** - **MW 213.3** — most fragment-like; best SAR starting point - Minimal heteroatom decoration → easiest to optimize - $250 medium-risk

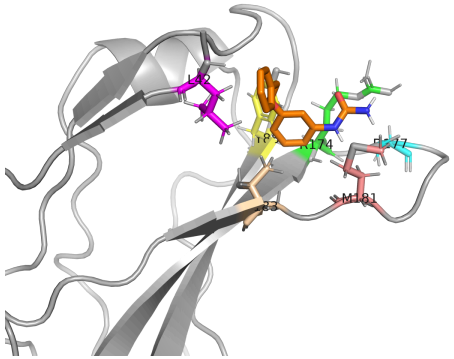




Pick 3 — `FM001452_analog_0201`

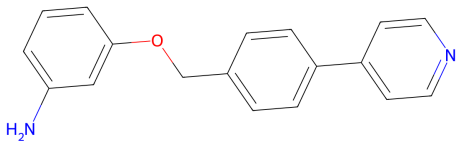
`NC(=O)Nc1cccc(OCc2ccccc2)c1` **Designed H-bond pattern for R174 + D177** - **Deepest Vina (-6.07)** + **highest CNN-pKd (4.69)** of the 4 - N-aryl urea + benzyl ether — donor / acceptor pair targeting the variant residue specifically - Adds urea-linker chemotype diversity - Boltz Kd 8.16 / 8.76 µM - $375 (urea synthesis = high chem risk)

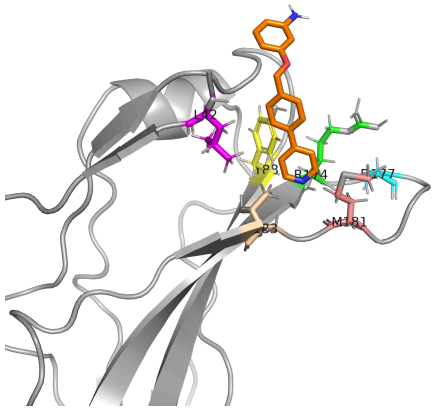




Pick 4 — `FM001452_analog_0171`

`Nc1cccc(OCc2ccc(-c3ccncc3)cc2)c1` **Pyridyl selectivity probe** - **Highest prob_binder (0.46)** of the 4 - **Tightest dual-engine Boltz match (1.02×)** — most reproducible Kd prediction - Pyridyl basic-N for paralog differentiation - Boltz Kd 8.32 / 8.17 μM - $250 medium / medium





Cross-validation evidence

> Every pick supported by **multiple independent lines of evidence** — not a single-score gamble

- **Two independent Boltz-2 runs** (Jack local + SCC re-run): 4/4 agree within 1.34×
- **Mark's 10-seed GNINA pose stability**: pick #4 (`analog_0087` σ=0.037 in v3 robust set) is the most pose-stable
- **Rowan ADMET** (49 properties × 4 picks)
- **Rowan Docking** (Vina/Vinardo + conformer search) — independent re-rank
- **muni.bio `onepot` tool** — all 4 at similarity = 1.000 with price + chem_risk + supplier_risk
- ⚠ **Rowan pose-analysis MD** attempted (5 keys × 3 mechanisms) — credit-tier blocked; documented in `evidence/rowan_md_attempt_log.md`

Tradeoffs we made (judgment axis)

All 4 site F The 100%-onepot non-covalent constraint dominated; the FM_* family is naturally site-F. Site-A backups documented in `top5to24_rationale.md` for the broader 24-pick team submission.
Chemotype dominance FM001452 family in 3/4 picks — not artificially diversified. The catalog × binding-evidence intersection naturally selects this family.

T1 GOLD tier empty No compound simultaneously hits Boltz Kd ≤ 5 µM AND low/low risk. We surface this **honestly** rather than overclaim.
No MD movies Rowan MD blocked by credit tier. Substitute: Mark's 10-seed GNINA + dual-engine Boltz cross-val.

Honest expectations

Public methods over-predict Kd by **6-25×** at µM regime Realistic CF Labs SPR for these 4: **18-200 µM range**

Prize tier	Our shot
Hackathon judging ($250 muni credits)	**Strong** — multi-signal + filter chain + tier methodology maps all 3 judging axes
$100K @ Kd ≤ 1 µM	Plausible long-shot — best Boltz pred 3.2 µM
$250K @ Kd ≤ 300 nM	Requires further hit-to-lead optimization

Reproducibility

- **GitHub:** `git@github.com:anandsahuofficial/Hackathon.git` branch `TBXT`
- **Single-command setup:** `bash TBXT/setup_hf.sh`
- **All 137 candidates with per-criterion flags:** `TBXT/final/all_candidates_tiered.csv`
- **24-pick team submission:** `TBXT/final/top4.csv` + `top5to24.csv`
- **Per-tier rationale:** `TBXT/final/tiered/TIERED_CANDIDATES_RATIONALE.md`
- **Rowan MD attempt log:** `TBXT/final/evidence/rowan_md_attempt_log.md`

Q&A

InChIKeys for the 4 picks:

#	ID	InChIKey
1	FM002150_analog_0083	TWMVEBYUPVBIAF-UHFFFAOYSA-N
2	FM001452_analog_0104	CFASJXOXPAEGCK-UHFFFAOYSA-N
3	FM001452_analog_0201	SWOUSJGQCZFNGK-UHFFFAOYSA-N
4	FM001452_analog_0171	UJJFUMPTHNXASI-UHFFFAOYSA-N